

```
In [1]: import numpy as np
import cv2
import os
import random
import matplotlib.pyplot as plt
import pickle
from sklearn.model_selection import train_test_split
from tensorflow.keras.models import Sequential
from tensorflow.keras.models import Model
import tensorflow as tf
from sklearn.metrics import classification_report, confusion_matrix
```

```
/opt/conda/lib/python3.10/site-packages/scipy/__init__.py:146: UserWarning: A NumPy
y version >=1.16.5 and <1.23.0 is required for this version of SciPy (detected ver
sion 1.23.5
  warnings.warn(f"A NumPy version >={np_minversion} and <{np_maxversion}")
```

```
In [2]: import tensorflow as tf
from tensorflow.keras import layers as L
import keras
from keras.layers import *
```

Data Preprocessing

```
In [3]: DIRECTORY = r'/kaggle/input/covid-ieee-train-test-split/COVID_IEEE Train Test Split'
CATEGORIES = ['covid', 'normal', 'virus']
IMG_SIZE = 224
train_data = []
```

```
In [4]: for category in CATEGORIES:
    folder = os.path.join(DIRECTORY, category)
    #print(folder)
    label = CATEGORIES.index(category)
    for img in os.listdir(folder):
        img_path = os.path.join(folder, img)
        #print(img_path)
        img_arr = cv2.imread(img_path)
        img_arr = cv2.resize(img_arr, (IMG_SIZE, IMG_SIZE))
        #plt.imshow(img_arr)
        #break
        train_data.append([img_arr, label])
```

```
In [5]: len(train_data)
```

```
Out[5]: 1457
```

```
In [6]: random.shuffle(train_data)
```

```
In [7]: X_train = []
y_train = []

for features, labels in train_data:
    X_train.append(features)
    y_train.append(labels)
```

```
In [8]: DIRECTORY = r'/kaggle/input/covid-ieee-train-test-split/COVID_IEEE Train Test Split'
CATEGORIES = ['covid', 'normal', 'virus']
```

```
IMG_SIZE = 224
test_data = []
```

```
In [9]: for category in CATEGORIES:
        folder = os.path.join(DIRECTORY, category)
        #print(folder)
        label = CATEGORIES.index(category)
        for img in os.listdir(folder):
            img_path = os.path.join(folder, img)
            #print(img_path)
            img_arr = cv2.imread(img_path)
            img_arr = cv2.resize(img_arr, (IMG_SIZE, IMG_SIZE))
            #plt.imshow(img_arr)
            #break
            test_data.append([img_arr, label])
```

```
In [10]: len(test_data)
```

```
Out[10]: 366
```

```
In [11]: X_test = []
        y_test = []

        for features, labels in test_data:
            X_test.append(features)
            y_test.append(labels)
```

```
In [12]: X_train = np.array(X_train)
        y_train = np.array(y_train)
        X_test = np.array(X_test)
        y_test = np.array(y_test)
```

```
In [13]: X_train.shape, X_test.shape, y_train.shape, y_test.shape
```

```
Out[13]: ((1457, 224, 224, 3), (366, 224, 224, 3), (1457,), (366,))
```

Model

```
In [14]: data_augmentation = keras.Sequential(
        [
            Normalization(),
            RandomFlip("horizontal"),
            RandomRotation(factor=0.02),
            RandomZoom(
                height_factor=0.2, width_factor=0.2
            ),
            RandomContrast(0.2),
            RandomBrightness(0.2)
        ],
        name="data_augmentation",
    )
```

```
In [15]: class SpatialAttentionModule(tf.keras.layers.Layer):
        def __init__(self, kernel_size=3):
            """
            paper: https://arxiv.org/abs/1807.06521
            code: https://gist.github.com/innat/99888fa8065ecbf3ae2b297e5c10db70
            """
            super(SpatialAttentionModule, self).__init__()
```

```

self.conv1 = tf.keras.layers.Conv2D(64, kernel_size=kernel_size,
                                     use_bias=False,
                                     kernel_initializer='he_normal',
                                     strides=1, padding='same',
                                     activation=tf.nn.relu)

self.conv2 = tf.keras.layers.Conv2D(32, kernel_size=kernel_size,
                                     use_bias=False,
                                     kernel_initializer='he_normal',
                                     strides=1, padding='same',
                                     activation=tf.nn.relu)

self.conv3 = tf.keras.layers.Conv2D(16, kernel_size=kernel_size,
                                     use_bias=False,
                                     kernel_initializer='he_normal',
                                     strides=1, padding='same',
                                     activation=tf.nn.relu)

self.conv4 = tf.keras.layers.Conv2D(1, kernel_size=kernel_size,
                                     use_bias=False,
                                     kernel_initializer='he_normal',
                                     strides=1, padding='same',
                                     activation=tf.math.sigmoid)

def call(self, inputs):
    avg_out = tf.reduce_mean(inputs, axis=3)
    max_out = tf.reduce_max(inputs, axis=3)
    x = tf.stack([avg_out, max_out], axis=3)
    x = self.conv1(x)
    x = self.conv2(x)
    x = self.conv3(x)
    return self.conv4(x)

class ChannelAttentionModule(tf.keras.layers.Layer):
    def __init__(self, ratio=8):
        """
        paper: https://arxiv.org/abs/1807.06521
        code: https://gist.github.com/innat/99888fa8065ecbf3ae2b297e5c10db70
        """
        super(ChannelAttentionModule, self).__init__()
        self.ratio = ratio
        self.gapavg = tf.keras.layers.GlobalAveragePooling2D()
        self.gmpmax = tf.keras.layers.GlobalMaxPooling2D()

    def build(self, input_shape):
        self.conv1 = tf.keras.layers.Conv2D(input_shape[-1]//self.ratio,
                                             kernel_size=1,
                                             strides=1, padding='same',
                                             use_bias=True, activation=tf.nn.relu)

        self.conv2 = tf.keras.layers.Conv2D(input_shape[-1],
                                             kernel_size=1,
                                             strides=1, padding='same',
                                             use_bias=True, activation=tf.nn.relu)
        super(ChannelAttentionModule, self).build(input_shape)

    def call(self, inputs):
        # compute gap and gmp pooling
        gapavg = self.gapavg(inputs)
        gmpmax = self.gmpmax(inputs)
        gapavg = tf.keras.layers.Reshape((1, 1, gapavg.shape[1]))(gapavg)
        gmpmax = tf.keras.layers.Reshape((1, 1, gmpmax.shape[1]))(gmpmax)
        # forward passing to the respected layers
        gapavg_out = self.conv2(self.conv1(gapavg))
        gmpmax_out = self.conv2(self.conv1(gmpmax))
        return tf.math.sigmoid(gapavg_out + gmpmax_out)

```

```
def get_output_shape_for(self, input_shape):
    return self.compute_output_shape(input_shape)

def compute_output_shape(self, input_shape):
    output_len = input_shape[3]
    return (input_shape[0], output_len)
```

```
In [16]: def casa_module(x):
    x_ca = ChannelAttentionModule()(x)
    x_ca_out = Multiply()([x_ca, x])
    x_sa = SpatialAttentionModule()(x)
    x_sa_out = Multiply()([x_sa, x])
    # Concatenate
    x_casa = Concatenate()([x_ca_out, x_sa_out])
    return x_casa
```

```
In [17]: def inverted_residual_block(inputs, num_filters, strides=1, expansion_ratio=1):
    # point-wise conv
    x = L.Conv2D(filters=expansion_ratio*inputs.shape[-1],
                  kernel_size=1,
                  padding='same',
                  use_bias=False)(inputs)
    x = L.BatchNormalization()(x)
    x = L.Activation('swish')(x)

    # depth-wise conv
    x = L.DepthwiseConv2D(kernel_size=3,
                           strides=strides,
                           padding='same',
                           use_bias=False)(x)
    x = L.BatchNormalization()(x)
    x = L.Activation('swish')(x)

    # point-wise conv
    x = L.Conv2D(filters=num_filters,
                  kernel_size=1,
                  padding='same',
                  use_bias=False)(x)
    x = L.BatchNormalization()(x)

    # Residual Connection
    if strides == 1 and (inputs.shape == x.shape):
        return L.Add()([inputs, x])
    return x
```

```
In [52]: inputs = Input((224,224,3))
    #x = ZeroPadding2D((3,3))(inputs)

    x = SeparableConv2D(64, (3,3), strides=(2,2))(inputs)
    x = BatchNormalization(axis=3)(x)
    x = Activation('relu')(x)

    x = SeparableConv2D(64, (3,3), activation='relu')(x)
    x = MaxPooling2D((2,2))(x)
    x = inverted_residual_block(x, 64, strides=1, expansion_ratio=2)

    x1 = MaxPooling2D((2,2))(x)
    x2 = casa_module(x)
    x2 = MaxPooling2D((2,2))(x)
    x = Concatenate()([x1, x2])
```

```
x = SeparableConv2D(128, (3,3), activation='relu')(x)
x = MaxPooling2D((2,2))(x)
x = SeparableConv2D(256, (3,3), activation='relu')(x)
x = MaxPooling2D((2,2))(x)
x = SeparableConv2D(256, (3,3), activation='relu')(x)
x = MaxPooling2D((2,2))(x)
```

```
In [53]: cnn = keras.Model(inputs, x)
```

```
In [54]: cnn.summary()
```

Model: "model_3"

Layer (type)	Output Shape	Param #	Connected to
=====			
input_8 (InputLayer)	[(None, 224, 224, 3)]	0	[]
separable_conv2d_17 (Separable Conv2D)	(None, 111, 111, 64)	283	['input_8[0][0]']
batch_normalization_13 (BatchN d_17[0][0]') ormalization)	(None, 111, 111, 64)	256	['separable_conv2
activation_10 (Activation) tion_13[0][0]']	(None, 111, 111, 64)	0	['batch_normaliza
separable_conv2d_18 (Separable [0][0]') Conv2D)	(None, 109, 109, 64)	4736	['activation_10
max_pooling2d_6 (MaxPooling2D) d_18[0][0]']	(None, 54, 54, 64)	0	['separable_conv2
conv2d_18 (Conv2D) [0][0]']	(None, 54, 54, 128)	8192	['max_pooling2d_6
batch_normalization_14 (BatchN [0]') ormalization)	(None, 54, 54, 128)	512	['conv2d_18[0]
activation_11 (Activation) tion_14[0][0]']	(None, 54, 54, 128)	0	['batch_normaliza
depthwise_conv2d_3 (DepthwiseC [0][0]') onv2D)	(None, 54, 54, 128)	1152	['activation_11
batch_normalization_15 (BatchN d_3[0][0]') ormalization)	(None, 54, 54, 128)	512	['depthwise_conv2
activation_12 (Activation) tion_15[0][0]']	(None, 54, 54, 128)	0	['batch_normaliza
conv2d_19 (Conv2D) [0][0]']	(None, 54, 54, 64)	8192	['activation_12
batch_normalization_16 (BatchN [0]') ormalization)	(None, 54, 54, 64)	256	['conv2d_19[0]
add_3 (Add) [0][0]',	(None, 54, 54, 64)	0	['max_pooling2d_6 'batch_normaliza
max_pooling2d_7 (MaxPooling2D)	(None, 27, 27, 64)	0	['add_3[0][0]']
max_pooling2d_8 (MaxPooling2D)	(None, 27, 27, 64)	0	['add_3[0][0]']

```

concatenate_7 (Concatenate)    (None, 27, 27, 128)  0      ['max_pooling2d_7
[0][0]',
                                'max_pooling2d_8
[0][0]']

separable_conv2d_19 (Separable  (None, 25, 25, 128)  17664   ['concatenate_7
[0][0]']
Conv2D)

max_pooling2d_9 (MaxPooling2D)  (None, 12, 12, 128)  0      ['separable_conv2
d_19[0][0]']

separable_conv2d_20 (Separable  (None, 10, 10, 256)  34176   ['max_pooling2d_9
[0][0]']
Conv2D)

max_pooling2d_10 (MaxPooling2D) (None, 5, 5, 256)    0      ['separable_conv2
d_20[0][0]']
)

separable_conv2d_21 (Separable  (None, 3, 3, 256)    68096   ['max_pooling2d_1
0[0][0]']
Conv2D)

max_pooling2d_11 (MaxPooling2D) (None, 1, 1, 256)    0      ['separable_conv2
d_21[0][0]']
)

=====
=====
Total params: 144,027
Trainable params: 143,259
Non-trainable params: 768

```

```

In [55]: model = tf.keras.Sequential([
    InputLayer((224,224,3)),
    data_augmentation,
    cnn,
    #Flatten(),
    GlobalAveragePooling2D(),
    #BatchNormalization(),
    Dropout(0.6),#0.75
    Dense(512, activation='relu'),
    Dropout(0.5),#0.5
    Dense(256, activation='relu'),
    #Dropout(0.2),#0.5
    Dense(128, activation='relu'),
    #Dense(64, activation='relu'),
    Dense(32, activation='relu'),
    Dense(3, activation='softmax')
])

```

```

In [56]: model.summary()

```

Model: "sequential_3"

Layer (type)	Output Shape	Param #
data_augmentation (Sequenti al)	(None, 224, 224, 3)	7
model_3 (Functional)	(None, 1, 1, 256)	144027
global_average_pooling2d_7 (GlobalAveragePooling2D)	(None, 256)	0
dropout_6 (Dropout)	(None, 256)	0
dense_15 (Dense)	(None, 512)	131584
dropout_7 (Dropout)	(None, 512)	0
dense_16 (Dense)	(None, 256)	131328
dense_17 (Dense)	(None, 128)	32896
dense_18 (Dense)	(None, 32)	4128
dense_19 (Dense)	(None, 3)	99
Total params: 444,069		
Trainable params: 443,294		
Non-trainable params: 775		

```
In [58]: # Compile the model
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

```
In [59]: class_weights = {0:1.1347,
                          1:0.9095,
                          2:0.9811}
```

```
In [60]: from keras.callbacks import ModelCheckpoint, ReduceLRonPlateau
```

```
In [61]: filepath = "/kaggle/working/Max_CASA_IRB_{epoch:02d}_{val_accuracy:.2f}.h5"
checkpoint = ModelCheckpoint(filepath, monitor="val_accuracy",
                             verbose=1, save_best_only=True,
                             mode='max')
reduce_lr = ReduceLRonPlateau(monitor='val_accuracy', factor=1e-1, patience=10,
                              verbose=1, min_lr = 0.00005)
callbacks = [checkpoint, reduce_lr]
```

```
In [62]: history = model.fit(X_train, y_train,
                             validation_data=[X_test, y_test],
                             class_weight=class_weights,
                             callbacks=callbacks,
                             epochs=250, batch_size=32)
```


Epoch 1/250
45/46 [=====>.] - ETA: 0s - loss: 1.0407 - accuracy: 0.4340
Epoch 1: val_accuracy improved from -inf to 0.29508, saving model to /kaggle/working/Max_CASA_IRB_01_0.30.h5
46/46 [=====] - 11s 82ms/step - loss: 1.0383 - accuracy: 0.4372 - val_loss: 1.1059 - val_accuracy: 0.2951 - lr: 0.0010
Epoch 2/250
45/46 [=====>.] - ETA: 0s - loss: 0.8393 - accuracy: 0.5938
Epoch 2: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 70ms/step - loss: 0.8370 - accuracy: 0.5944 - val_loss: 1.7568 - val_accuracy: 0.2951 - lr: 0.0010
Epoch 3/250
45/46 [=====>.] - ETA: 0s - loss: 0.6429 - accuracy: 0.7021
Epoch 3: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 73ms/step - loss: 0.6405 - accuracy: 0.7035 - val_loss: 1.8954 - val_accuracy: 0.2951 - lr: 0.0010
Epoch 4/250
45/46 [=====>.] - ETA: 0s - loss: 0.4892 - accuracy: 0.8049
Epoch 4: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 74ms/step - loss: 0.4868 - accuracy: 0.8058 - val_loss: 2.0137 - val_accuracy: 0.2951 - lr: 0.0010
Epoch 5/250
45/46 [=====>.] - ETA: 0s - loss: 0.3779 - accuracy: 0.8625
Epoch 5: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 70ms/step - loss: 0.3836 - accuracy: 0.8614 - val_loss: 1.9586 - val_accuracy: 0.2951 - lr: 0.0010
Epoch 6/250
45/46 [=====>.] - ETA: 0s - loss: 0.3333 - accuracy: 0.8833
Epoch 6: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 71ms/step - loss: 0.3307 - accuracy: 0.8847 - val_loss: 1.9924 - val_accuracy: 0.2951 - lr: 0.0010
Epoch 7/250
45/46 [=====>.] - ETA: 0s - loss: 0.3530 - accuracy: 0.8785
Epoch 7: val_accuracy improved from 0.29508 to 0.30328, saving model to /kaggle/working/Max_CASA_IRB_07_0.30.h5
46/46 [=====] - 3s 73ms/step - loss: 0.3530 - accuracy: 0.8785 - val_loss: 1.4215 - val_accuracy: 0.3033 - lr: 0.0010
Epoch 8/250
45/46 [=====>.] - ETA: 0s - loss: 0.3474 - accuracy: 0.8854
Epoch 8: val_accuracy improved from 0.30328 to 0.40710, saving model to /kaggle/working/Max_CASA_IRB_08_0.41.h5
46/46 [=====] - 3s 73ms/step - loss: 0.3472 - accuracy: 0.8854 - val_loss: 1.3833 - val_accuracy: 0.4071 - lr: 0.0010
Epoch 9/250
45/46 [=====>.] - ETA: 0s - loss: 0.2975 - accuracy: 0.8931
Epoch 9: val_accuracy improved from 0.40710 to 0.64754, saving model to /kaggle/working/Max_CASA_IRB_09_0.65.h5
46/46 [=====] - 3s 74ms/step - loss: 0.3008 - accuracy: 0.8916 - val_loss: 0.7641 - val_accuracy: 0.6475 - lr: 0.0010
Epoch 10/250
45/46 [=====>.] - ETA: 0s - loss: 0.2419 - accuracy: 0.9208
Epoch 10: val_accuracy improved from 0.64754 to 0.68033, saving model to /kaggle/working/Max_CASA_IRB_10_0.68.h5
46/46 [=====] - 3s 74ms/step - loss: 0.2408 - accuracy: 0.9211 - val_loss: 0.7715 - val_accuracy: 0.6803 - lr: 0.0010
Epoch 11/250
45/46 [=====>.] - ETA: 0s - loss: 0.2536 - accuracy: 0.9056
Epoch 11: val_accuracy improved from 0.68033 to 0.92623, saving model to /kaggle/working/Max_CASA_IRB_11_0.93.h5
46/46 [=====] - 3s 73ms/step - loss: 0.2523 - accuracy: 0.9060 - val_loss: 0.3621 - val_accuracy: 0.9262 - lr: 0.0010
Epoch 12/250
45/46 [=====>.] - ETA: 0s - loss: 0.2470 - accuracy: 0.9187
Epoch 12: val_accuracy did not improve from 0.92623

```
46/46 [=====] - 3s 71ms/step - loss: 0.2492 - accuracy:
0.9190 - val_loss: 1.2817 - val_accuracy: 0.5027 - lr: 0.0010
Epoch 13/250
45/46 [=====>.] - ETA: 0s - loss: 0.2588 - accuracy: 0.9132
Epoch 13: val_accuracy did not improve from 0.92623
46/46 [=====] - 3s 73ms/step - loss: 0.2592 - accuracy:
0.9135 - val_loss: 0.3688 - val_accuracy: 0.8770 - lr: 0.0010
Epoch 14/250
45/46 [=====>.] - ETA: 0s - loss: 0.2539 - accuracy: 0.9187
Epoch 14: val_accuracy improved from 0.92623 to 0.93716, saving model to /kaggle/w
orking/Max_CASA_IRB_14_0.94.h5
46/46 [=====] - 4s 78ms/step - loss: 0.2518 - accuracy:
0.9197 - val_loss: 0.1950 - val_accuracy: 0.9372 - lr: 0.0010
Epoch 15/250
45/46 [=====>.] - ETA: 0s - loss: 0.2393 - accuracy: 0.9194
Epoch 15: val_accuracy did not improve from 0.93716
46/46 [=====] - 3s 74ms/step - loss: 0.2397 - accuracy:
0.9190 - val_loss: 0.3702 - val_accuracy: 0.8661 - lr: 0.0010
Epoch 16/250
45/46 [=====>.] - ETA: 0s - loss: 0.2403 - accuracy: 0.9222
Epoch 16: val_accuracy did not improve from 0.93716
46/46 [=====] - 3s 71ms/step - loss: 0.2418 - accuracy:
0.9218 - val_loss: 0.2442 - val_accuracy: 0.9262 - lr: 0.0010
Epoch 17/250
45/46 [=====>.] - ETA: 0s - loss: 0.2350 - accuracy: 0.9243
Epoch 17: val_accuracy did not improve from 0.93716
46/46 [=====] - 3s 73ms/step - loss: 0.2335 - accuracy:
0.9252 - val_loss: 0.3026 - val_accuracy: 0.8962 - lr: 0.0010
Epoch 18/250
45/46 [=====>.] - ETA: 0s - loss: 0.2310 - accuracy: 0.9250
Epoch 18: val_accuracy improved from 0.93716 to 0.94536, saving model to /kaggle/w
orking/Max_CASA_IRB_18_0.95.h5
46/46 [=====] - 3s 76ms/step - loss: 0.2295 - accuracy:
0.9259 - val_loss: 0.1884 - val_accuracy: 0.9454 - lr: 0.0010
Epoch 19/250
45/46 [=====>.] - ETA: 0s - loss: 0.2294 - accuracy: 0.9278
Epoch 19: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.2325 - accuracy:
0.9266 - val_loss: 0.2041 - val_accuracy: 0.9372 - lr: 0.0010
Epoch 20/250
45/46 [=====>.] - ETA: 0s - loss: 0.2224 - accuracy: 0.9299
Epoch 20: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.2212 - accuracy:
0.9307 - val_loss: 0.2761 - val_accuracy: 0.9044 - lr: 0.0010
Epoch 21/250
45/46 [=====>.] - ETA: 0s - loss: 0.2031 - accuracy: 0.9438
Epoch 21: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.2016 - accuracy:
0.9444 - val_loss: 0.3397 - val_accuracy: 0.8907 - lr: 0.0010
Epoch 22/250
45/46 [=====>.] - ETA: 0s - loss: 0.2287 - accuracy: 0.9257
Epoch 22: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.2301 - accuracy:
0.9245 - val_loss: 0.4111 - val_accuracy: 0.8497 - lr: 0.0010
Epoch 23/250
45/46 [=====>.] - ETA: 0s - loss: 0.2087 - accuracy: 0.9299
Epoch 23: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.2088 - accuracy:
0.9300 - val_loss: 0.2433 - val_accuracy: 0.9208 - lr: 0.0010
Epoch 24/250
45/46 [=====>.] - ETA: 0s - loss: 0.2281 - accuracy: 0.9153
Epoch 24: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.2271 - accuracy:
0.9156 - val_loss: 0.1919 - val_accuracy: 0.9426 - lr: 0.0010
```

Epoch 25/250
45/46 [=====>.] - ETA: 0s - loss: 0.2310 - accuracy: 0.9264
Epoch 25: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.2288 - accuracy: 0.9272 - val_loss: 0.2636 - val_accuracy: 0.9153 - lr: 0.0010
Epoch 26/250
45/46 [=====>.] - ETA: 0s - loss: 0.1935 - accuracy: 0.9340
Epoch 26: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1917 - accuracy: 0.9348 - val_loss: 0.2646 - val_accuracy: 0.9126 - lr: 0.0010
Epoch 27/250
45/46 [=====>.] - ETA: 0s - loss: 0.1976 - accuracy: 0.9382
Epoch 27: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1963 - accuracy: 0.9389 - val_loss: 0.2184 - val_accuracy: 0.9372 - lr: 0.0010
Epoch 28/250
45/46 [=====>.] - ETA: 0s - loss: 0.1893 - accuracy: 0.9375
Epoch 28: val_accuracy did not improve from 0.94536

Epoch 28: ReduceLRonPlateau reducing learning rate to 0.00010000000474974513.
46/46 [=====] - 3s 72ms/step - loss: 0.1901 - accuracy: 0.9369 - val_loss: 0.2809 - val_accuracy: 0.9044 - lr: 0.0010
Epoch 29/250
45/46 [=====>.] - ETA: 0s - loss: 0.1704 - accuracy: 0.9431
Epoch 29: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1692 - accuracy: 0.9437 - val_loss: 0.2167 - val_accuracy: 0.9317 - lr: 1.0000e-04
Epoch 30/250
45/46 [=====>.] - ETA: 0s - loss: 0.1621 - accuracy: 0.9451
Epoch 30: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1619 - accuracy: 0.9451 - val_loss: 0.2279 - val_accuracy: 0.9235 - lr: 1.0000e-04
Epoch 31/250
45/46 [=====>.] - ETA: 0s - loss: 0.1482 - accuracy: 0.9514
Epoch 31: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1503 - accuracy: 0.9506 - val_loss: 0.2214 - val_accuracy: 0.9262 - lr: 1.0000e-04
Epoch 32/250
45/46 [=====>.] - ETA: 0s - loss: 0.1418 - accuracy: 0.9549
Epoch 32: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1411 - accuracy: 0.9554 - val_loss: 0.2418 - val_accuracy: 0.9262 - lr: 1.0000e-04
Epoch 33/250
45/46 [=====>.] - ETA: 0s - loss: 0.1418 - accuracy: 0.9507
Epoch 33: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.1425 - accuracy: 0.9506 - val_loss: 0.2565 - val_accuracy: 0.9235 - lr: 1.0000e-04
Epoch 34/250
45/46 [=====>.] - ETA: 0s - loss: 0.1491 - accuracy: 0.9500
Epoch 34: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1477 - accuracy: 0.9506 - val_loss: 0.2359 - val_accuracy: 0.9262 - lr: 1.0000e-04
Epoch 35/250
45/46 [=====>.] - ETA: 0s - loss: 0.1370 - accuracy: 0.9514
Epoch 35: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1370 - accuracy: 0.9513 - val_loss: 0.2211 - val_accuracy: 0.9290 - lr: 1.0000e-04
Epoch 36/250
45/46 [=====>.] - ETA: 0s - loss: 0.1517 - accuracy: 0.9486
Epoch 36: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1503 - accuracy: 0.9492 - val_loss: 0.2450 - val_accuracy: 0.9262 - lr: 1.0000e-04
Epoch 37/250
45/46 [=====>.] - ETA: 0s - loss: 0.1286 - accuracy: 0.9549

Epoch 37: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1307 - accuracy: 0.9547 - val_loss: 0.2354 - val_accuracy: 0.9235 - lr: 1.0000e-04
Epoch 38/250
45/46 [=====>.] - ETA: 0s - loss: 0.1364 - accuracy: 0.9549
Epoch 38: val_accuracy did not improve from 0.94536

Epoch 38: ReduceLROnPlateau reducing learning rate to 5e-05.
46/46 [=====] - 3s 70ms/step - loss: 0.1396 - accuracy: 0.9540 - val_loss: 0.2353 - val_accuracy: 0.9262 - lr: 1.0000e-04
Epoch 39/250
45/46 [=====>.] - ETA: 0s - loss: 0.1376 - accuracy: 0.9521
Epoch 39: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1378 - accuracy: 0.9520 - val_loss: 0.2337 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 40/250
45/46 [=====>.] - ETA: 0s - loss: 0.1417 - accuracy: 0.9479
Epoch 40: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1454 - accuracy: 0.9472 - val_loss: 0.2394 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 41/250
45/46 [=====>.] - ETA: 0s - loss: 0.1469 - accuracy: 0.9549
Epoch 41: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1475 - accuracy: 0.9540 - val_loss: 0.2456 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 42/250
45/46 [=====>.] - ETA: 0s - loss: 0.1324 - accuracy: 0.9590
Epoch 42: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1318 - accuracy: 0.9588 - val_loss: 0.2364 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 43/250
45/46 [=====>.] - ETA: 0s - loss: 0.1400 - accuracy: 0.9528
Epoch 43: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1388 - accuracy: 0.9533 - val_loss: 0.2297 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 44/250
45/46 [=====>.] - ETA: 0s - loss: 0.1473 - accuracy: 0.9556
Epoch 44: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1467 - accuracy: 0.9554 - val_loss: 0.2497 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 45/250
45/46 [=====>.] - ETA: 0s - loss: 0.1294 - accuracy: 0.9535
Epoch 45: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1295 - accuracy: 0.9533 - val_loss: 0.2354 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 46/250
45/46 [=====>.] - ETA: 0s - loss: 0.1427 - accuracy: 0.9535
Epoch 46: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1437 - accuracy: 0.9533 - val_loss: 0.2538 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 47/250
45/46 [=====>.] - ETA: 0s - loss: 0.1429 - accuracy: 0.9556
Epoch 47: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1420 - accuracy: 0.9554 - val_loss: 0.2604 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 48/250
45/46 [=====>.] - ETA: 0s - loss: 0.1398 - accuracy: 0.9563
Epoch 48: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1392 - accuracy: 0.9561 - val_loss: 0.2691 - val_accuracy: 0.9098 - lr: 5.0000e-05
Epoch 49/250
45/46 [=====>.] - ETA: 0s - loss: 0.1427 - accuracy: 0.9528
Epoch 49: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1418 - accuracy:

0.9533 - val_loss: 0.2516 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 50/250
45/46 [=====>.] - ETA: 0s - loss: 0.1367 - accuracy: 0.9569
Epoch 50: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1376 - accuracy: 0.9561 - val_loss: 0.2323 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 51/250
45/46 [=====>.] - ETA: 0s - loss: 0.1286 - accuracy: 0.9569
Epoch 51: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1273 - accuracy: 0.9574 - val_loss: 0.2284 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 52/250
45/46 [=====>.] - ETA: 0s - loss: 0.1324 - accuracy: 0.9549
Epoch 52: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1332 - accuracy: 0.9547 - val_loss: 0.2220 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 53/250
45/46 [=====>.] - ETA: 0s - loss: 0.1187 - accuracy: 0.9604
Epoch 53: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1184 - accuracy: 0.9602 - val_loss: 0.2119 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 54/250
46/46 [=====] - ETA: 0s - loss: 0.1418 - accuracy: 0.9533
Epoch 54: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1418 - accuracy: 0.9533 - val_loss: 0.2235 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 55/250
45/46 [=====>.] - ETA: 0s - loss: 0.1313 - accuracy: 0.9563
Epoch 55: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1313 - accuracy: 0.9561 - val_loss: 0.2306 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 56/250
45/46 [=====>.] - ETA: 0s - loss: 0.1308 - accuracy: 0.9563
Epoch 56: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1298 - accuracy: 0.9568 - val_loss: 0.2464 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 57/250
45/46 [=====>.] - ETA: 0s - loss: 0.1220 - accuracy: 0.9590
Epoch 57: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1212 - accuracy: 0.9595 - val_loss: 0.2596 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 58/250
45/46 [=====>.] - ETA: 0s - loss: 0.1162 - accuracy: 0.9611
Epoch 58: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1156 - accuracy: 0.9616 - val_loss: 0.2624 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 59/250
45/46 [=====>.] - ETA: 0s - loss: 0.1259 - accuracy: 0.9569
Epoch 59: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1300 - accuracy: 0.9561 - val_loss: 0.2428 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 60/250
45/46 [=====>.] - ETA: 0s - loss: 0.1224 - accuracy: 0.9618
Epoch 60: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1214 - accuracy: 0.9623 - val_loss: 0.2358 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 61/250
45/46 [=====>.] - ETA: 0s - loss: 0.1122 - accuracy: 0.9611
Epoch 61: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1155 - accuracy: 0.9595 - val_loss: 0.2218 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 62/250
45/46 [=====>.] - ETA: 0s - loss: 0.1100 - accuracy: 0.9625
Epoch 62: val_accuracy did not improve from 0.94536

46/46 [=====] - 3s 72ms/step - loss: 0.1093 - accuracy: 0.9629 - val_loss: 0.2357 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 63/250
45/46 [=====>.] - ETA: 0s - loss: 0.1186 - accuracy: 0.9590
Epoch 63: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1175 - accuracy: 0.9595 - val_loss: 0.2446 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 64/250
45/46 [=====>.] - ETA: 0s - loss: 0.1270 - accuracy: 0.9604
Epoch 64: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.1287 - accuracy: 0.9602 - val_loss: 0.2358 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 65/250
45/46 [=====>.] - ETA: 0s - loss: 0.1221 - accuracy: 0.9583
Epoch 65: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.1210 - accuracy: 0.9588 - val_loss: 0.2359 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 66/250
45/46 [=====>.] - ETA: 0s - loss: 0.1349 - accuracy: 0.9514
Epoch 66: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1352 - accuracy: 0.9513 - val_loss: 0.2422 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 67/250
45/46 [=====>.] - ETA: 0s - loss: 0.1155 - accuracy: 0.9639
Epoch 67: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1143 - accuracy: 0.9643 - val_loss: 0.2068 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 68/250
45/46 [=====>.] - ETA: 0s - loss: 0.1214 - accuracy: 0.9556
Epoch 68: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1235 - accuracy: 0.9547 - val_loss: 0.2094 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 69/250
45/46 [=====>.] - ETA: 0s - loss: 0.1191 - accuracy: 0.9576
Epoch 69: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1187 - accuracy: 0.9574 - val_loss: 0.2122 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 70/250
45/46 [=====>.] - ETA: 0s - loss: 0.1256 - accuracy: 0.9590
Epoch 70: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1262 - accuracy: 0.9588 - val_loss: 0.2122 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 71/250
45/46 [=====>.] - ETA: 0s - loss: 0.1307 - accuracy: 0.9563
Epoch 71: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1344 - accuracy: 0.9554 - val_loss: 0.2140 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 72/250
45/46 [=====>.] - ETA: 0s - loss: 0.1283 - accuracy: 0.9563
Epoch 72: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1287 - accuracy: 0.9554 - val_loss: 0.2334 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 73/250
45/46 [=====>.] - ETA: 0s - loss: 0.1220 - accuracy: 0.9569
Epoch 73: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1214 - accuracy: 0.9574 - val_loss: 0.2259 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 74/250
45/46 [=====>.] - ETA: 0s - loss: 0.1277 - accuracy: 0.9611
Epoch 74: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1280 - accuracy: 0.9609 - val_loss: 0.2247 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 75/250
45/46 [=====>.] - ETA: 0s - loss: 0.1189 - accuracy: 0.9604

Epoch 75: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1192 - accuracy: 0.9602 - val_loss: 0.1966 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 76/250
45/46 [=====>.] - ETA: 0s - loss: 0.1267 - accuracy: 0.9563
Epoch 76: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1258 - accuracy: 0.9568 - val_loss: 0.2032 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 77/250
45/46 [=====>.] - ETA: 0s - loss: 0.1095 - accuracy: 0.9653
Epoch 77: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1101 - accuracy: 0.9650 - val_loss: 0.2131 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 78/250
45/46 [=====>.] - ETA: 0s - loss: 0.1263 - accuracy: 0.9542
Epoch 78: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1294 - accuracy: 0.9533 - val_loss: 0.2072 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 79/250
45/46 [=====>.] - ETA: 0s - loss: 0.1250 - accuracy: 0.9542
Epoch 79: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1242 - accuracy: 0.9547 - val_loss: 0.2031 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 80/250
45/46 [=====>.] - ETA: 0s - loss: 0.1127 - accuracy: 0.9611
Epoch 80: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1124 - accuracy: 0.9609 - val_loss: 0.2348 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 81/250
45/46 [=====>.] - ETA: 0s - loss: 0.1154 - accuracy: 0.9625
Epoch 81: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1157 - accuracy: 0.9623 - val_loss: 0.2191 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 82/250
45/46 [=====>.] - ETA: 0s - loss: 0.1209 - accuracy: 0.9597
Epoch 82: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1202 - accuracy: 0.9602 - val_loss: 0.2251 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 83/250
45/46 [=====>.] - ETA: 0s - loss: 0.0923 - accuracy: 0.9694
Epoch 83: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.0927 - accuracy: 0.9691 - val_loss: 0.2337 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 84/250
45/46 [=====>.] - ETA: 0s - loss: 0.1218 - accuracy: 0.9583
Epoch 84: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1228 - accuracy: 0.9581 - val_loss: 0.2159 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 85/250
45/46 [=====>.] - ETA: 0s - loss: 0.1254 - accuracy: 0.9625
Epoch 85: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1241 - accuracy: 0.9629 - val_loss: 0.2370 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 86/250
45/46 [=====>.] - ETA: 0s - loss: 0.1110 - accuracy: 0.9597
Epoch 86: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1105 - accuracy: 0.9602 - val_loss: 0.2109 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 87/250
45/46 [=====>.] - ETA: 0s - loss: 0.1107 - accuracy: 0.9646
Epoch 87: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1118 - accuracy: 0.9643 - val_loss: 0.1918 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 88/250

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45/46 [=====>.] - ETA: 0s - loss: 0.1269 - accuracy: 0.9583
Epoch 88: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1263 - accuracy:
0.9581 - val_loss: 0.2092 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 89/250
45/46 [=====>.] - ETA: 0s - loss: 0.1223 - accuracy: 0.9611
Epoch 89: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1211 - accuracy:
0.9616 - val_loss: 0.2452 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 90/250
45/46 [=====>.] - ETA: 0s - loss: 0.1339 - accuracy: 0.9535
Epoch 90: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1335 - accuracy:
0.9533 - val_loss: 0.2332 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 91/250
45/46 [=====>.] - ETA: 0s - loss: 0.1112 - accuracy: 0.9653
Epoch 91: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1104 - accuracy:
0.9657 - val_loss: 0.2131 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 92/250
45/46 [=====>.] - ETA: 0s - loss: 0.1084 - accuracy: 0.9667
Epoch 92: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1083 - accuracy:
0.9664 - val_loss: 0.1996 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 93/250
45/46 [=====>.] - ETA: 0s - loss: 0.1240 - accuracy: 0.9556
Epoch 93: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1242 - accuracy:
0.9554 - val_loss: 0.2041 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 94/250
45/46 [=====>.] - ETA: 0s - loss: 0.1165 - accuracy: 0.9604
Epoch 94: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1157 - accuracy:
0.9602 - val_loss: 0.2064 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 95/250
45/46 [=====>.] - ETA: 0s - loss: 0.1272 - accuracy: 0.9563
Epoch 95: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1319 - accuracy:
0.9540 - val_loss: 0.2078 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 96/250
45/46 [=====>.] - ETA: 0s - loss: 0.1130 - accuracy: 0.9611
Epoch 96: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1121 - accuracy:
0.9616 - val_loss: 0.2118 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 97/250
45/46 [=====>.] - ETA: 0s - loss: 0.1123 - accuracy: 0.9653
Epoch 97: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1115 - accuracy:
0.9657 - val_loss: 0.2269 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 98/250
45/46 [=====>.] - ETA: 0s - loss: 0.1155 - accuracy: 0.9646
Epoch 98: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1172 - accuracy:
0.9643 - val_loss: 0.2118 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 99/250
45/46 [=====>.] - ETA: 0s - loss: 0.1088 - accuracy: 0.9625
Epoch 99: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1079 - accuracy:
0.9629 - val_loss: 0.2197 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 100/250
45/46 [=====>.] - ETA: 0s - loss: 0.1202 - accuracy: 0.9611
Epoch 100: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 76ms/step - loss: 0.1199 - accuracy:
0.9609 - val_loss: 0.2234 - val_accuracy: 0.9262 - lr: 5.0000e-05
```


Epoch 101/250
45/46 [=====>.] - ETA: 0s - loss: 0.1154 - accuracy: 0.9667
Epoch 101: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.1154 - accuracy: 0.9664 - val_loss: 0.2056 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 102/250
45/46 [=====>.] - ETA: 0s - loss: 0.1167 - accuracy: 0.9556
Epoch 102: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1157 - accuracy: 0.9561 - val_loss: 0.1940 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 103/250
45/46 [=====>.] - ETA: 0s - loss: 0.1093 - accuracy: 0.9625
Epoch 103: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1083 - accuracy: 0.9629 - val_loss: 0.2055 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 104/250
45/46 [=====>.] - ETA: 0s - loss: 0.1029 - accuracy: 0.9639
Epoch 104: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1020 - accuracy: 0.9643 - val_loss: 0.2171 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 105/250
45/46 [=====>.] - ETA: 0s - loss: 0.1055 - accuracy: 0.9688
Epoch 105: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1074 - accuracy: 0.9684 - val_loss: 0.2106 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 106/250
45/46 [=====>.] - ETA: 0s - loss: 0.1251 - accuracy: 0.9597
Epoch 106: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1242 - accuracy: 0.9602 - val_loss: 0.1992 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 107/250
45/46 [=====>.] - ETA: 0s - loss: 0.1139 - accuracy: 0.9611
Epoch 107: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1127 - accuracy: 0.9616 - val_loss: 0.2012 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 108/250
45/46 [=====>.] - ETA: 0s - loss: 0.1046 - accuracy: 0.9639
Epoch 108: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1038 - accuracy: 0.9643 - val_loss: 0.1966 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 109/250
45/46 [=====>.] - ETA: 0s - loss: 0.1129 - accuracy: 0.9618
Epoch 109: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1117 - accuracy: 0.9623 - val_loss: 0.1894 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 110/250
45/46 [=====>.] - ETA: 0s - loss: 0.1132 - accuracy: 0.9681
Epoch 110: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1143 - accuracy: 0.9677 - val_loss: 0.1966 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 111/250
45/46 [=====>.] - ETA: 0s - loss: 0.1170 - accuracy: 0.9618
Epoch 111: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1159 - accuracy: 0.9623 - val_loss: 0.1931 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 112/250
45/46 [=====>.] - ETA: 0s - loss: 0.1096 - accuracy: 0.9639
Epoch 112: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1095 - accuracy: 0.9636 - val_loss: 0.2126 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 113/250
45/46 [=====>.] - ETA: 0s - loss: 0.1208 - accuracy: 0.9590
Epoch 113: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1196 - accuracy:

0.9595 - val_loss: 0.2265 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 114/250
45/46 [=====>.] - ETA: 0s - loss: 0.1143 - accuracy: 0.9653
Epoch 114: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1136 - accuracy: 0.9657 - val_loss: 0.1949 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 115/250
45/46 [=====>.] - ETA: 0s - loss: 0.1203 - accuracy: 0.9576
Epoch 115: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1196 - accuracy: 0.9574 - val_loss: 0.1910 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 116/250
45/46 [=====>.] - ETA: 0s - loss: 0.0947 - accuracy: 0.9681
Epoch 116: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.0976 - accuracy: 0.9671 - val_loss: 0.2049 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 117/250
45/46 [=====>.] - ETA: 0s - loss: 0.1153 - accuracy: 0.9632
Epoch 117: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1158 - accuracy: 0.9623 - val_loss: 0.2002 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 118/250
45/46 [=====>.] - ETA: 0s - loss: 0.0958 - accuracy: 0.9674
Epoch 118: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.0949 - accuracy: 0.9677 - val_loss: 0.2039 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 119/250
45/46 [=====>.] - ETA: 0s - loss: 0.0960 - accuracy: 0.9715
Epoch 119: val_accuracy did not improve from 0.94536
46/46 [=====] - 4s 77ms/step - loss: 0.0961 - accuracy: 0.9712 - val_loss: 0.1971 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 120/250
45/46 [=====>.] - ETA: 0s - loss: 0.0884 - accuracy: 0.9681
Epoch 120: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.0877 - accuracy: 0.9684 - val_loss: 0.1854 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 121/250
45/46 [=====>.] - ETA: 0s - loss: 0.1027 - accuracy: 0.9646
Epoch 121: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1051 - accuracy: 0.9636 - val_loss: 0.2009 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 122/250
45/46 [=====>.] - ETA: 0s - loss: 0.0979 - accuracy: 0.9694
Epoch 122: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.0988 - accuracy: 0.9691 - val_loss: 0.1811 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 123/250
45/46 [=====>.] - ETA: 0s - loss: 0.1033 - accuracy: 0.9618
Epoch 123: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.1049 - accuracy: 0.9616 - val_loss: 0.1938 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 124/250
45/46 [=====>.] - ETA: 0s - loss: 0.0989 - accuracy: 0.9688
Epoch 124: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 73ms/step - loss: 0.0992 - accuracy: 0.9684 - val_loss: 0.1929 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 125/250
45/46 [=====>.] - ETA: 0s - loss: 0.1135 - accuracy: 0.9660
Epoch 125: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 70ms/step - loss: 0.1124 - accuracy: 0.9664 - val_loss: 0.1917 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 126/250
45/46 [=====>.] - ETA: 0s - loss: 0.1111 - accuracy: 0.9639
Epoch 126: val_accuracy improved from 0.94536 to 0.94809, saving model to /kaggle/

working/Max_CASA_IRB_126_0.95.h5

46/46 [=====] - 3s 73ms/step - loss: 0.1112 - accuracy: 0.9636 - val_loss: 0.1958 - val_accuracy: 0.9481 - lr: 5.0000e-05
Epoch 127/250
45/46 [=====>.] - ETA: 0s - loss: 0.1062 - accuracy: 0.9604
Epoch 127: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.1054 - accuracy: 0.9609 - val_loss: 0.1947 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 128/250
45/46 [=====>.] - ETA: 0s - loss: 0.1074 - accuracy: 0.9625
Epoch 128: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.1064 - accuracy: 0.9629 - val_loss: 0.1888 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 129/250
45/46 [=====>.] - ETA: 0s - loss: 0.1078 - accuracy: 0.9604
Epoch 129: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 75ms/step - loss: 0.1069 - accuracy: 0.9609 - val_loss: 0.1906 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 130/250
45/46 [=====>.] - ETA: 0s - loss: 0.1030 - accuracy: 0.9590
Epoch 130: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.1019 - accuracy: 0.9595 - val_loss: 0.1942 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 131/250
45/46 [=====>.] - ETA: 0s - loss: 0.1112 - accuracy: 0.9639
Epoch 131: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.1104 - accuracy: 0.9643 - val_loss: 0.1898 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 132/250
45/46 [=====>.] - ETA: 0s - loss: 0.0939 - accuracy: 0.9674
Epoch 132: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0942 - accuracy: 0.9671 - val_loss: 0.2014 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 133/250
45/46 [=====>.] - ETA: 0s - loss: 0.1171 - accuracy: 0.9639
Epoch 133: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.1166 - accuracy: 0.9643 - val_loss: 0.2086 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 134/250
45/46 [=====>.] - ETA: 0s - loss: 0.1101 - accuracy: 0.9590
Epoch 134: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.1137 - accuracy: 0.9574 - val_loss: 0.2004 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 135/250
45/46 [=====>.] - ETA: 0s - loss: 0.1087 - accuracy: 0.9653
Epoch 135: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.1087 - accuracy: 0.9650 - val_loss: 0.2195 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 136/250
45/46 [=====>.] - ETA: 0s - loss: 0.1024 - accuracy: 0.9674
Epoch 136: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.1015 - accuracy: 0.9677 - val_loss: 0.2123 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 137/250
45/46 [=====>.] - ETA: 0s - loss: 0.0961 - accuracy: 0.9722
Epoch 137: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.0954 - accuracy: 0.9725 - val_loss: 0.1925 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 138/250
45/46 [=====>.] - ETA: 0s - loss: 0.0967 - accuracy: 0.9660
Epoch 138: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 75ms/step - loss: 0.0958 - accuracy: 0.9664 - val_loss: 0.2108 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 139/250

45/46 [=====>.] - ETA: 0s - loss: 0.0987 - accuracy: 0.9618
Epoch 139: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0981 - accuracy: 0.9623 - val_loss: 0.2043 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 140/250
45/46 [=====>.] - ETA: 0s - loss: 0.0894 - accuracy: 0.9701
Epoch 140: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0887 - accuracy: 0.9705 - val_loss: 0.2148 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 141/250
45/46 [=====>.] - ETA: 0s - loss: 0.1051 - accuracy: 0.9625
Epoch 141: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.1041 - accuracy: 0.9629 - val_loss: 0.2069 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 142/250
45/46 [=====>.] - ETA: 0s - loss: 0.1037 - accuracy: 0.9660
Epoch 142: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.1026 - accuracy: 0.9664 - val_loss: 0.2106 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 143/250
45/46 [=====>.] - ETA: 0s - loss: 0.0991 - accuracy: 0.9646
Epoch 143: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.1006 - accuracy: 0.9636 - val_loss: 0.1986 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 144/250
45/46 [=====>.] - ETA: 0s - loss: 0.1014 - accuracy: 0.9653
Epoch 144: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.1005 - accuracy: 0.9657 - val_loss: 0.2008 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 145/250
45/46 [=====>.] - ETA: 0s - loss: 0.0968 - accuracy: 0.9701
Epoch 145: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0957 - accuracy: 0.9705 - val_loss: 0.2100 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 146/250
45/46 [=====>.] - ETA: 0s - loss: 0.0991 - accuracy: 0.9646
Epoch 146: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0994 - accuracy: 0.9643 - val_loss: 0.2161 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 147/250
45/46 [=====>.] - ETA: 0s - loss: 0.1060 - accuracy: 0.9653
Epoch 147: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.1052 - accuracy: 0.9657 - val_loss: 0.2071 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 148/250
45/46 [=====>.] - ETA: 0s - loss: 0.1080 - accuracy: 0.9688
Epoch 148: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.1076 - accuracy: 0.9684 - val_loss: 0.2077 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 149/250
45/46 [=====>.] - ETA: 0s - loss: 0.0922 - accuracy: 0.9674
Epoch 149: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0914 - accuracy: 0.9677 - val_loss: 0.2184 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 150/250
45/46 [=====>.] - ETA: 0s - loss: 0.0946 - accuracy: 0.9646
Epoch 150: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0968 - accuracy: 0.9643 - val_loss: 0.2011 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 151/250
45/46 [=====>.] - ETA: 0s - loss: 0.0959 - accuracy: 0.9701
Epoch 151: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0952 - accuracy: 0.9705 - val_loss: 0.2000 - val_accuracy: 0.9262 - lr: 5.0000e-05

Epoch 152/250
45/46 [=====>.] - ETA: 0s - loss: 0.0941 - accuracy: 0.9701
Epoch 152: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0931 - accuracy: 0.9705 - val_loss: 0.2124 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 153/250
45/46 [=====>.] - ETA: 0s - loss: 0.0871 - accuracy: 0.9715
Epoch 153: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0864 - accuracy: 0.9719 - val_loss: 0.2017 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 154/250
45/46 [=====>.] - ETA: 0s - loss: 0.0962 - accuracy: 0.9639
Epoch 154: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0962 - accuracy: 0.9636 - val_loss: 0.1989 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 155/250
45/46 [=====>.] - ETA: 0s - loss: 0.0787 - accuracy: 0.9771
Epoch 155: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0782 - accuracy: 0.9774 - val_loss: 0.1809 - val_accuracy: 0.9481 - lr: 5.0000e-05
Epoch 156/250
45/46 [=====>.] - ETA: 0s - loss: 0.0968 - accuracy: 0.9653
Epoch 156: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0979 - accuracy: 0.9650 - val_loss: 0.1820 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 157/250
45/46 [=====>.] - ETA: 0s - loss: 0.1001 - accuracy: 0.9660
Epoch 157: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0992 - accuracy: 0.9664 - val_loss: 0.1743 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 158/250
45/46 [=====>.] - ETA: 0s - loss: 0.0986 - accuracy: 0.9667
Epoch 158: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.0979 - accuracy: 0.9671 - val_loss: 0.1786 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 159/250
45/46 [=====>.] - ETA: 0s - loss: 0.0985 - accuracy: 0.9646
Epoch 159: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0976 - accuracy: 0.9650 - val_loss: 0.1803 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 160/250
45/46 [=====>.] - ETA: 0s - loss: 0.1016 - accuracy: 0.9646
Epoch 160: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.1016 - accuracy: 0.9643 - val_loss: 0.1916 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 161/250
45/46 [=====>.] - ETA: 0s - loss: 0.0855 - accuracy: 0.9743
Epoch 161: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0848 - accuracy: 0.9746 - val_loss: 0.1989 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 162/250
45/46 [=====>.] - ETA: 0s - loss: 0.0950 - accuracy: 0.9694
Epoch 162: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0953 - accuracy: 0.9691 - val_loss: 0.1917 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 163/250
45/46 [=====>.] - ETA: 0s - loss: 0.0845 - accuracy: 0.9736
Epoch 163: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0837 - accuracy: 0.9739 - val_loss: 0.1996 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 164/250
45/46 [=====>.] - ETA: 0s - loss: 0.0958 - accuracy: 0.9625
Epoch 164: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0948 - accuracy:

0.9629 - val_loss: 0.2043 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 165/250
45/46 [=====>.] - ETA: 0s - loss: 0.1051 - accuracy: 0.9660
Epoch 165: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.1042 - accuracy: 0.9664 - val_loss: 0.2633 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 166/250
45/46 [=====>.] - ETA: 0s - loss: 0.0980 - accuracy: 0.9688
Epoch 166: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0977 - accuracy: 0.9684 - val_loss: 0.2458 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 167/250
45/46 [=====>.] - ETA: 0s - loss: 0.0911 - accuracy: 0.9688
Epoch 167: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 75ms/step - loss: 0.0951 - accuracy: 0.9677 - val_loss: 0.2253 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 168/250
45/46 [=====>.] - ETA: 0s - loss: 0.0879 - accuracy: 0.9715
Epoch 168: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0880 - accuracy: 0.9712 - val_loss: 0.2031 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 169/250
45/46 [=====>.] - ETA: 0s - loss: 0.0911 - accuracy: 0.9681
Epoch 169: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0907 - accuracy: 0.9677 - val_loss: 0.1900 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 170/250
45/46 [=====>.] - ETA: 0s - loss: 0.0853 - accuracy: 0.9674
Epoch 170: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0864 - accuracy: 0.9671 - val_loss: 0.2185 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 171/250
45/46 [=====>.] - ETA: 0s - loss: 0.0824 - accuracy: 0.9729
Epoch 171: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0821 - accuracy: 0.9725 - val_loss: 0.1887 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 172/250
45/46 [=====>.] - ETA: 0s - loss: 0.1003 - accuracy: 0.9688
Epoch 172: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0997 - accuracy: 0.9691 - val_loss: 0.2008 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 173/250
45/46 [=====>.] - ETA: 0s - loss: 0.0964 - accuracy: 0.9681
Epoch 173: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.0955 - accuracy: 0.9684 - val_loss: 0.1970 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 174/250
45/46 [=====>.] - ETA: 0s - loss: 0.0844 - accuracy: 0.9708
Epoch 174: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.0836 - accuracy: 0.9712 - val_loss: 0.2017 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 175/250
45/46 [=====>.] - ETA: 0s - loss: 0.0908 - accuracy: 0.9715
Epoch 175: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0902 - accuracy: 0.9719 - val_loss: 0.1750 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 176/250
45/46 [=====>.] - ETA: 0s - loss: 0.1000 - accuracy: 0.9715
Epoch 176: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0994 - accuracy: 0.9719 - val_loss: 0.1770 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 177/250
45/46 [=====>.] - ETA: 0s - loss: 0.0895 - accuracy: 0.9688
Epoch 177: val_accuracy did not improve from 0.94809

46/46 [=====] - 3s 75ms/step - loss: 0.0895 - accuracy: 0.9684 - val_loss: 0.2086 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 178/250
45/46 [=====>.] - ETA: 0s - loss: 0.0941 - accuracy: 0.9715
Epoch 178: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0931 - accuracy: 0.9719 - val_loss: 0.2189 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 179/250
45/46 [=====>.] - ETA: 0s - loss: 0.0949 - accuracy: 0.9660
Epoch 179: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0939 - accuracy: 0.9664 - val_loss: 0.1959 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 180/250
45/46 [=====>.] - ETA: 0s - loss: 0.0895 - accuracy: 0.9681
Epoch 180: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0886 - accuracy: 0.9684 - val_loss: 0.1890 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 181/250
45/46 [=====>.] - ETA: 0s - loss: 0.0789 - accuracy: 0.9715
Epoch 181: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0785 - accuracy: 0.9719 - val_loss: 0.2110 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 182/250
45/46 [=====>.] - ETA: 0s - loss: 0.1016 - accuracy: 0.9660
Epoch 182: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.1007 - accuracy: 0.9664 - val_loss: 0.2208 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 183/250
45/46 [=====>.] - ETA: 0s - loss: 0.0949 - accuracy: 0.9646
Epoch 183: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0965 - accuracy: 0.9643 - val_loss: 0.1999 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 184/250
45/46 [=====>.] - ETA: 0s - loss: 0.0824 - accuracy: 0.9701
Epoch 184: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0819 - accuracy: 0.9705 - val_loss: 0.2195 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 185/250
45/46 [=====>.] - ETA: 0s - loss: 0.0950 - accuracy: 0.9639
Epoch 185: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0947 - accuracy: 0.9643 - val_loss: 0.1936 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 186/250
45/46 [=====>.] - ETA: 0s - loss: 0.0869 - accuracy: 0.9667
Epoch 186: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.0864 - accuracy: 0.9671 - val_loss: 0.2147 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 187/250
45/46 [=====>.] - ETA: 0s - loss: 0.0869 - accuracy: 0.9701
Epoch 187: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0883 - accuracy: 0.9698 - val_loss: 0.2378 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 188/250
45/46 [=====>.] - ETA: 0s - loss: 0.0846 - accuracy: 0.9743
Epoch 188: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0874 - accuracy: 0.9739 - val_loss: 0.2038 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 189/250
45/46 [=====>.] - ETA: 0s - loss: 0.0910 - accuracy: 0.9694
Epoch 189: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0906 - accuracy: 0.9698 - val_loss: 0.2242 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 190/250
45/46 [=====>.] - ETA: 0s - loss: 0.0930 - accuracy: 0.9694

Epoch 190: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0924 - accuracy: 0.9698 - val_loss: 0.2144 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 191/250
45/46 [=====>.] - ETA: 0s - loss: 0.0916 - accuracy: 0.9708
Epoch 191: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0912 - accuracy: 0.9712 - val_loss: 0.2190 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 192/250
45/46 [=====>.] - ETA: 0s - loss: 0.0856 - accuracy: 0.9708
Epoch 192: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.0848 - accuracy: 0.9712 - val_loss: 0.2188 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 193/250
45/46 [=====>.] - ETA: 0s - loss: 0.0742 - accuracy: 0.9757
Epoch 193: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0734 - accuracy: 0.9760 - val_loss: 0.2030 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 194/250
45/46 [=====>.] - ETA: 0s - loss: 0.0773 - accuracy: 0.9722
Epoch 194: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0766 - accuracy: 0.9725 - val_loss: 0.2009 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 195/250
45/46 [=====>.] - ETA: 0s - loss: 0.0868 - accuracy: 0.9722
Epoch 195: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0924 - accuracy: 0.9712 - val_loss: 0.1997 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 196/250
45/46 [=====>.] - ETA: 0s - loss: 0.0834 - accuracy: 0.9708
Epoch 196: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0826 - accuracy: 0.9712 - val_loss: 0.2169 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 197/250
45/46 [=====>.] - ETA: 0s - loss: 0.0810 - accuracy: 0.9715
Epoch 197: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0807 - accuracy: 0.9712 - val_loss: 0.2076 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 198/250
45/46 [=====>.] - ETA: 0s - loss: 0.0764 - accuracy: 0.9722
Epoch 198: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0772 - accuracy: 0.9712 - val_loss: 0.2001 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 199/250
45/46 [=====>.] - ETA: 0s - loss: 0.0863 - accuracy: 0.9722
Epoch 199: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0880 - accuracy: 0.9719 - val_loss: 0.2189 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 200/250
45/46 [=====>.] - ETA: 0s - loss: 0.0939 - accuracy: 0.9660
Epoch 200: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0929 - accuracy: 0.9664 - val_loss: 0.2040 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 201/250
45/46 [=====>.] - ETA: 0s - loss: 0.0848 - accuracy: 0.9708
Epoch 201: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0840 - accuracy: 0.9712 - val_loss: 0.1898 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 202/250
45/46 [=====>.] - ETA: 0s - loss: 0.0759 - accuracy: 0.9715
Epoch 202: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0763 - accuracy: 0.9712 - val_loss: 0.1981 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 203/250


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45/46 [=====>.] - ETA: 0s - loss: 0.0770 - accuracy: 0.9708
Epoch 203: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0785 - accuracy:
0.9698 - val_loss: 0.2065 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 204/250
45/46 [=====>.] - ETA: 0s - loss: 0.0831 - accuracy: 0.9694
Epoch 204: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0827 - accuracy:
0.9698 - val_loss: 0.2239 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 205/250
45/46 [=====>.] - ETA: 0s - loss: 0.0909 - accuracy: 0.9688
Epoch 205: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0912 - accuracy:
0.9684 - val_loss: 0.1983 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 206/250
45/46 [=====>.] - ETA: 0s - loss: 0.0850 - accuracy: 0.9736
Epoch 206: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0847 - accuracy:
0.9732 - val_loss: 0.1846 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 207/250
45/46 [=====>.] - ETA: 0s - loss: 0.0771 - accuracy: 0.9736
Epoch 207: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 73ms/step - loss: 0.0811 - accuracy:
0.9719 - val_loss: 0.2061 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 208/250
45/46 [=====>.] - ETA: 0s - loss: 0.0772 - accuracy: 0.9715
Epoch 208: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0764 - accuracy:
0.9719 - val_loss: 0.1847 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 209/250
45/46 [=====>.] - ETA: 0s - loss: 0.0876 - accuracy: 0.9701
Epoch 209: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 71ms/step - loss: 0.0867 - accuracy:
0.9705 - val_loss: 0.1945 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 210/250
45/46 [=====>.] - ETA: 0s - loss: 0.0977 - accuracy: 0.9667
Epoch 210: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0968 - accuracy:
0.9671 - val_loss: 0.1936 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 211/250
45/46 [=====>.] - ETA: 0s - loss: 0.0930 - accuracy: 0.9708
Epoch 211: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0922 - accuracy:
0.9712 - val_loss: 0.1991 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 212/250
45/46 [=====>.] - ETA: 0s - loss: 0.0794 - accuracy: 0.9729
Epoch 212: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0786 - accuracy:
0.9732 - val_loss: 0.2185 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 213/250
45/46 [=====>.] - ETA: 0s - loss: 0.0855 - accuracy: 0.9736
Epoch 213: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0862 - accuracy:
0.9732 - val_loss: 0.2129 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 214/250
45/46 [=====>.] - ETA: 0s - loss: 0.0912 - accuracy: 0.9667
Epoch 214: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0906 - accuracy:
0.9671 - val_loss: 0.1840 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 215/250
45/46 [=====>.] - ETA: 0s - loss: 0.0862 - accuracy: 0.9701
Epoch 215: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 75ms/step - loss: 0.0855 - accuracy:
0.9705 - val_loss: 0.2292 - val_accuracy: 0.9262 - lr: 5.0000e-05
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Epoch 216/250
45/46 [=====>.] - ETA: 0s - loss: 0.0797 - accuracy: 0.9750
Epoch 216: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.0790 - accuracy: 0.9753 - val_loss: 0.2047 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 217/250
45/46 [=====>.] - ETA: 0s - loss: 0.0851 - accuracy: 0.9701
Epoch 217: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 70ms/step - loss: 0.0860 - accuracy: 0.9698 - val_loss: 0.1785 - val_accuracy: 0.9481 - lr: 5.0000e-05
Epoch 218/250
45/46 [=====>.] - ETA: 0s - loss: 0.0934 - accuracy: 0.9660
Epoch 218: val_accuracy improved from 0.94809 to 0.95082, saving model to /kaggle/working/Max_CASA_IRB_218_0.95.h5
46/46 [=====] - 3s 73ms/step - loss: 0.0923 - accuracy: 0.9664 - val_loss: 0.1689 - val_accuracy: 0.9508 - lr: 5.0000e-05
Epoch 219/250
45/46 [=====>.] - ETA: 0s - loss: 0.0840 - accuracy: 0.9708
Epoch 219: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0837 - accuracy: 0.9712 - val_loss: 0.1672 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 220/250
45/46 [=====>.] - ETA: 0s - loss: 0.0787 - accuracy: 0.9729
Epoch 220: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 70ms/step - loss: 0.0789 - accuracy: 0.9732 - val_loss: 0.1915 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 221/250
45/46 [=====>.] - ETA: 0s - loss: 0.0774 - accuracy: 0.9736
Epoch 221: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 71ms/step - loss: 0.0767 - accuracy: 0.9739 - val_loss: 0.1891 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 222/250
45/46 [=====>.] - ETA: 0s - loss: 0.0895 - accuracy: 0.9694
Epoch 222: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 70ms/step - loss: 0.0886 - accuracy: 0.9698 - val_loss: 0.1964 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 223/250
45/46 [=====>.] - ETA: 0s - loss: 0.0831 - accuracy: 0.9701
Epoch 223: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 70ms/step - loss: 0.0827 - accuracy: 0.9705 - val_loss: 0.1714 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 224/250
45/46 [=====>.] - ETA: 0s - loss: 0.0993 - accuracy: 0.9625
Epoch 224: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 73ms/step - loss: 0.0982 - accuracy: 0.9629 - val_loss: 0.1858 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 225/250
45/46 [=====>.] - ETA: 0s - loss: 0.0742 - accuracy: 0.9743
Epoch 225: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0735 - accuracy: 0.9746 - val_loss: 0.1821 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 226/250
45/46 [=====>.] - ETA: 0s - loss: 0.0750 - accuracy: 0.9736
Epoch 226: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0765 - accuracy: 0.9732 - val_loss: 0.2021 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 227/250
45/46 [=====>.] - ETA: 0s - loss: 0.0933 - accuracy: 0.9646
Epoch 227: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0924 - accuracy: 0.9650 - val_loss: 0.1851 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 228/250
45/46 [=====>.] - ETA: 0s - loss: 0.0970 - accuracy: 0.9667
Epoch 228: val_accuracy did not improve from 0.95082

46/46 [=====] - 3s 72ms/step - loss: 0.0960 - accuracy: 0.9671 - val_loss: 0.1975 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 229/250
45/46 [=====>.] - ETA: 0s - loss: 0.0716 - accuracy: 0.9743
Epoch 229: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 73ms/step - loss: 0.0711 - accuracy: 0.9746 - val_loss: 0.2213 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 230/250
45/46 [=====>.] - ETA: 0s - loss: 0.0817 - accuracy: 0.9688
Epoch 230: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 73ms/step - loss: 0.0838 - accuracy: 0.9684 - val_loss: 0.2015 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 231/250
45/46 [=====>.] - ETA: 0s - loss: 0.0789 - accuracy: 0.9701
Epoch 231: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0799 - accuracy: 0.9698 - val_loss: 0.2241 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 232/250
45/46 [=====>.] - ETA: 0s - loss: 0.0832 - accuracy: 0.9722
Epoch 232: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0823 - accuracy: 0.9725 - val_loss: 0.2131 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 233/250
45/46 [=====>.] - ETA: 0s - loss: 0.0815 - accuracy: 0.9688
Epoch 233: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 73ms/step - loss: 0.0813 - accuracy: 0.9691 - val_loss: 0.2056 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 234/250
45/46 [=====>.] - ETA: 0s - loss: 0.0981 - accuracy: 0.9701
Epoch 234: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 74ms/step - loss: 0.0996 - accuracy: 0.9691 - val_loss: 0.1993 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 235/250
45/46 [=====>.] - ETA: 0s - loss: 0.0777 - accuracy: 0.9750
Epoch 235: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 70ms/step - loss: 0.0772 - accuracy: 0.9753 - val_loss: 0.2098 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 236/250
45/46 [=====>.] - ETA: 0s - loss: 0.0802 - accuracy: 0.9708
Epoch 236: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 71ms/step - loss: 0.0794 - accuracy: 0.9712 - val_loss: 0.2209 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 237/250
45/46 [=====>.] - ETA: 0s - loss: 0.0741 - accuracy: 0.9764
Epoch 237: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0736 - accuracy: 0.9767 - val_loss: 0.2245 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 238/250
45/46 [=====>.] - ETA: 0s - loss: 0.0821 - accuracy: 0.9701
Epoch 238: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 70ms/step - loss: 0.0813 - accuracy: 0.9705 - val_loss: 0.2169 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 239/250
45/46 [=====>.] - ETA: 0s - loss: 0.0651 - accuracy: 0.9764
Epoch 239: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 73ms/step - loss: 0.0649 - accuracy: 0.9767 - val_loss: 0.2252 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 240/250
45/46 [=====>.] - ETA: 0s - loss: 0.0880 - accuracy: 0.9750
Epoch 240: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0871 - accuracy: 0.9753 - val_loss: 0.2205 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 241/250
45/46 [=====>.] - ETA: 0s - loss: 0.0782 - accuracy: 0.9722

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Epoch 241: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 70ms/step - loss: 0.0775 - accuracy:
0.9725 - val_loss: 0.2199 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 242/250
45/46 [=====>.] - ETA: 0s - loss: 0.0858 - accuracy: 0.9653
Epoch 242: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 73ms/step - loss: 0.0850 - accuracy:
0.9657 - val_loss: 0.2354 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 243/250
45/46 [=====>.] - ETA: 0s - loss: 0.0685 - accuracy: 0.9771
Epoch 243: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0680 - accuracy:
0.9774 - val_loss: 0.2357 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 244/250
45/46 [=====>.] - ETA: 0s - loss: 0.0796 - accuracy: 0.9722
Epoch 244: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 75ms/step - loss: 0.0793 - accuracy:
0.9719 - val_loss: 0.2132 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 245/250
45/46 [=====>.] - ETA: 0s - loss: 0.0818 - accuracy: 0.9743
Epoch 245: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 71ms/step - loss: 0.0813 - accuracy:
0.9746 - val_loss: 0.1904 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 246/250
45/46 [=====>.] - ETA: 0s - loss: 0.0728 - accuracy: 0.9708
Epoch 246: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 74ms/step - loss: 0.0733 - accuracy:
0.9705 - val_loss: 0.1836 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 247/250
45/46 [=====>.] - ETA: 0s - loss: 0.0786 - accuracy: 0.9715
Epoch 247: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 71ms/step - loss: 0.0779 - accuracy:
0.9719 - val_loss: 0.1832 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 248/250
45/46 [=====>.] - ETA: 0s - loss: 0.0705 - accuracy: 0.9778
Epoch 248: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 73ms/step - loss: 0.0721 - accuracy:
0.9774 - val_loss: 0.1852 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 249/250
45/46 [=====>.] - ETA: 0s - loss: 0.0812 - accuracy: 0.9722
Epoch 249: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 73ms/step - loss: 0.0808 - accuracy:
0.9725 - val_loss: 0.1935 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 250/250
45/46 [=====>.] - ETA: 0s - loss: 0.0813 - accuracy: 0.9701
Epoch 250: val_accuracy did not improve from 0.95082
46/46 [=====] - 3s 72ms/step - loss: 0.0808 - accuracy:
0.9705 - val_loss: 0.1859 - val_accuracy: 0.9481 - lr: 5.0000e-05

```

```

In [63]: print(history.history.keys())

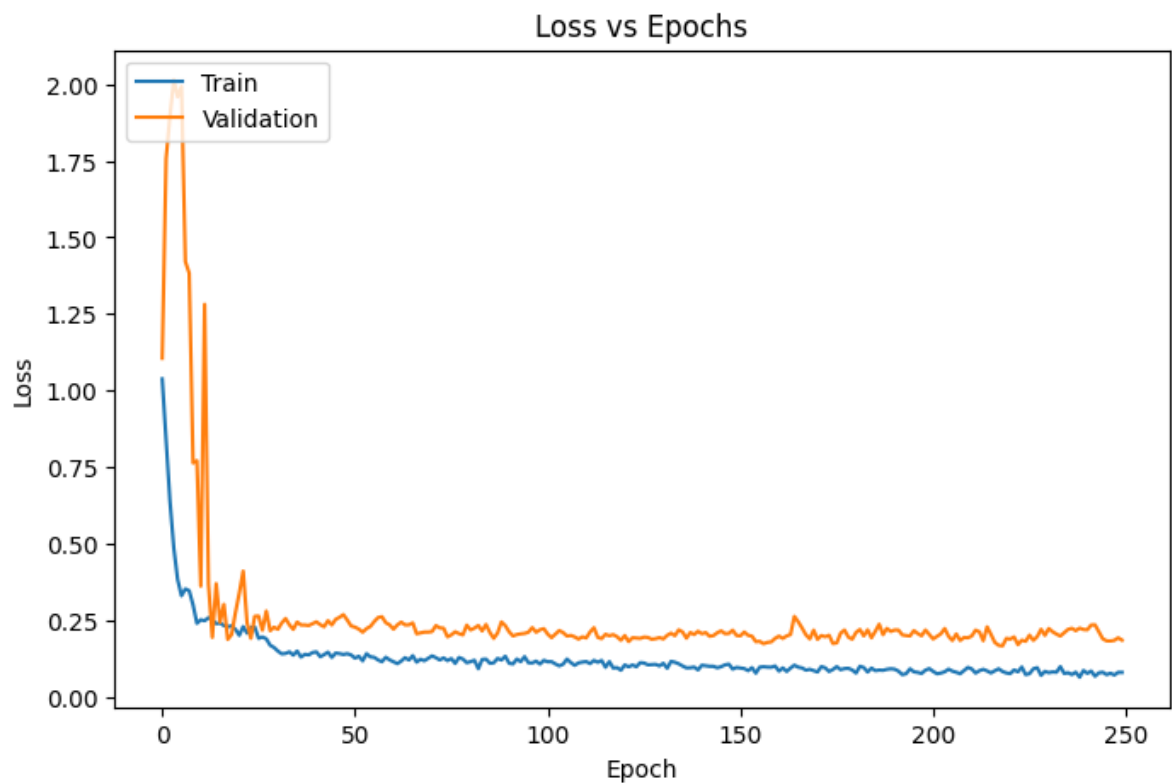
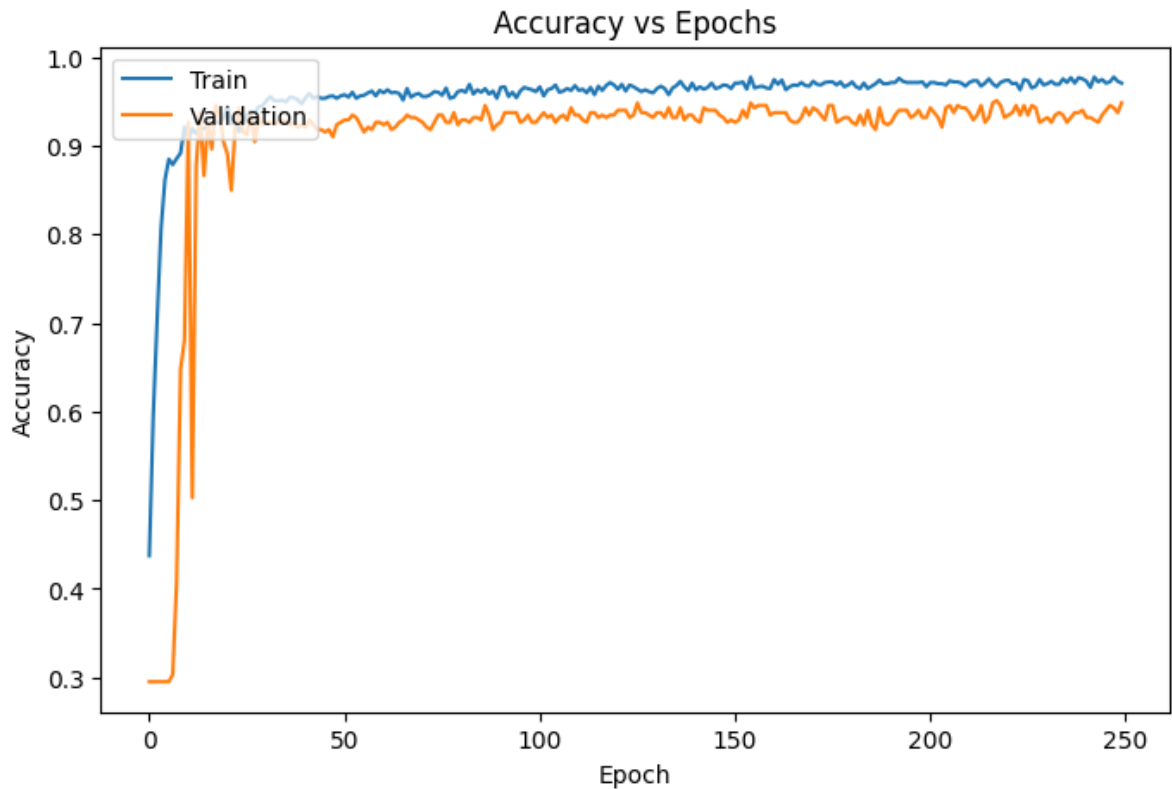
# "Accuracy"
plt.figure(figsize=(8, 5))
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('Accuracy vs Epochs')
plt.ylabel('Accuracy')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'], loc='upper left')
plt.savefig('acc_95_MaxCaSaIRB.png', dpi=1200)
plt.show()

# "Loss"
plt.figure(figsize=(8, 5))

```

```
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Loss vs Epochs')
plt.ylabel('Loss')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'], loc='upper left')
plt.savefig('loss_95_MaxCaSaIRB.png', dpi=1200)
plt.show()
```

dict_keys(['loss', 'accuracy', 'val_loss', 'val_accuracy', 'lr'])



Performance Evaluation

[illegible]

```
In [65]: re_model.evaluate(X_test, y_test)
```

```
12/12 [=====] - 1s 22ms/step - loss: 0.1689 - accuracy: 0.9508
```

```
Out[65]: [0.16893461346626282, 0.9508196711540222]
```

```
In [66]: y_pred = re_model.predict(X_test)
```

12/12 [=====] - 0s 20ms/step

```
In [67]: y_pred = np.argmax(y_pred, axis=1)
          y_pred
```

```
Out[67]: array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 1,
1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1,
1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2,
2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 1, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 0, 2, 2])
```

```
In [68]: y_true = y_test
```

```
In [69]: print(classification_report(y_true, y_pred))
```

	precision	recall	f1-score	support
0	0.97	0.97	0.97	108
1	0.95	0.93	0.94	134
2	0.94	0.95	0.94	124
accuracy			0.95	366
macro avg	0.95	0.95	0.95	366
weighted avg	0.95	0.95	0.95	366

```
In [70]: print(confusion_matrix(y_true, y_pred))
```

$$\begin{bmatrix} 105 & 2 & 1 \\ 2 & 125 & 7 \\ 1 & 5 & 118 \end{bmatrix}$$

In []: